CardioTwin X

An Adaptive, Explainable Cardiovascular Digital Twin for Transparent Cardiac Monitoring

Theme: Swasth & Samavesh Bharat	**Sub-theme:** Healthcare Innovation	**Track:** Product / Technology Solution

1. Overview

CardioTwin X is a software-based cardiac digital twin: a virtual model of a person's cardiovascular system that updates itself as new heart rate, blood pressure and other measurements arrive. It lets users see how the system is changing, understand why in plain language, see how reliable the result is, and simulate how the system might respond to changing conditions. We do not claim to have invented cardiac digital twins. Our aim is a practical, transparent and interactive implementation that addresses specific usability and reliability gaps.

2. Problem Statement

Most cardiac dashboards only display current readings, and most AI tools give a risk score without explaining it or stating how confident they are. This creates three problems:

- **Missed change:** readings are judged against generic ranges, so meaningful changes in an individual's own cardiovascular state can go unnoticed.
- **Low trust:** unexplained alerts, or confident scores built on poor or missing data, are hard for doctors, students and patients to trust.
- **No safe exploration:** users cannot easily test how stress, exercise, salt intake or arterial stiffness would affect blood pressure and heart rate.

The gap matters in India, where hypertension and heart disease are widespread, are often detected late, and specialist access is limited outside major cities.

3. Proposed Solution and Key Features

Feature	What it does
Dynamic Patient Twin	Updates the virtual patient's state with every new reading and tracks its own uncertainty.
Divergence Engine	Compares observed data with the twin's expected state, normalized by model uncertainty, to flag statistically meaningful deviations.
Explainable Alerts	Shows which signals and which physiological parameters (e.g. vessel resistance) explain an alert.
Uncertainty Awareness	Displays prediction bands and a data reliability level; says so when data is insufficient instead of showing a confident result.
What-If Simulator	Lets users change exercise, stress, salt intake or arterial stiffness and compare predicted response with the baseline.
Validation Dashboard	Reports prediction error, uncertainty calibration and detection performance against reference data.
Accessible Interface	Interactive 3D heart view, trend charts and a patient timeline in a simple dashboard.

4. How It Works

- A simplified 3-element Windkessel model represents arterial pressure using vessel resistance, arterial compliance and stroke volume.
- For each new reading, the twin predicts what it expects, compares it with the real value and adjusts its parameters using a Kalman filter.
- The Divergence Engine scores the difference and marks the state Normal, Watch or Significant, requiring persistence to limit false alarms. An ensemble of twins and an Isolation Forest provide uncertainty bands and a second opinion.
- The explanation module reports parameter shifts versus the patient's baseline, with a confidence label and data reliability level.
- What-if scenarios change selected parameters and show the predicted pressure waveform and vitals against the baseline.

5. Innovation

- Transparent comparison of observed and expected cardiovascular state, normalized by model uncertainty.
- Explanations in terms of physiological parameters, not only feature-importance scores.
- Honest handling of weak data: the system states when predictions are unreliable.
- A built-in validation dashboard so accuracy and calibration are visible to users.
- Lightweight software that runs on ordinary computers with no special hardware.

6. Technology Stack

Python, NumPy, SciPy (physiological model and ODE solving), scikit-learn (anomaly detection), Streamlit (interface), Plotly (charts and 3D visualization). Data: synthetic patient data generated by the project, with planned use of appropriately licensed public ECG and blood pressure datasets (for example PTB-XL).

7. Current Stage and Evidence

A working software prototype has been built and tested on synthetic data. Automated tests check that the default model gives physiologically reasonable pressure (about 120/80 mmHg), that the estimator recovers known parameters, and that the divergence score rises after an injected anomaly. These results show the method works as designed; they do not demonstrate clinical accuracy.

8. Beneficiaries and Impact

- **Patients and families:** clearer understanding of their readings and how lifestyle factors affect them.
- **Primary care doctors:** an explainable second view that states its uncertainty.
- **Students and educators:** an interactive tool for learning cardiovascular physiology.
- Near-term impact is awareness, trust in AI and health education. Clinical impact would require proper validation first.

9. Roadmap

Phase	Focus
Phase 1 (0-3 months)	Complete validation on synthetic and public datasets; release as a free educational and research web app.
Phase 2 (3-6 months)	Pilot with a medical or engineering college; collect usability feedback from students and faculty.
Phase 3 (6-12 months)	Add wearable or home BP device input and a clinician view for supervised pilots, with ethical approval.

10. Estimated Cost

- Prototype: close to zero beyond existing hardware, using open-source tools.
- 6-month pilot: approximately Rs. 1.5 to 3 lakh (hosting, sensor integration, usability study).
- Running cost: low, a few thousand rupees per month for pilot hosting.
- Clinical validation and regulatory work are not included and would need separate funding.

11. Limitations and Responsible Use

- The prototype uses synthetic data and a simplified physiological model; it is not clinically validated and must not be used for diagnosis or treatment decisions.
- Because synthetic data comes from a similar model, validation on public real-world datasets is essential before any stronger claim.
- What-if outputs are illustrative simulations, not medical advice.
- Any use with real patients needs ethical approval, data protection safeguards and clinical review.

12. Support Needed

- Mentorship from a cardiologist or physiologist to review the model and validation approach.
- Guidance on medical device regulation and data protection.
- Access to anonymized, appropriately licensed clinical datasets.
- Funding for sensor integration and a small pilot, and a partner hospital or college for usability testing.